

Ohm Patel

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EDUCATION

Brown University

M.S., Computer Science

Sc.B., Mechanical Engineering · GPA 3.96 / 4.00

Providence, RI

Sep 2027 – Expected May 2028

Sep 2023 – May 2027

Coursework: Data Structures & Algorithms, Computer Systems, Object-Oriented Programming (Java), Differential Equations

SKILLS

Product: PRD authoring, user stories and requirements, roadmap planning, competitive analysis, market sizing, adoption metrics, spec handoff to engineering

Design thinking and agile: Empathy interviews, journey mapping, ideation sessions · sprint retrospectives, daily standups, iterative delivery and staged releases · Jira, GitHub

AI and ML: LLM APIs, retrieval-augmented generation, model and tool evaluation, output guardrails and responsible AI practices, embedding models and vector stores

Technical: Python, SQL, TypeScript, Java · FastAPI, Next.js, serverless, REST APIs · Docker, CI/CD, Pytest, Git · Pandas, NumPy, Plotly

Communication: PowerPoint, Word, Excel · technical writing, editorial direction, presenting to non-technical stakeholders

PRODUCT PROJECTS

Cairn: Course Copilot

2026

Solo build · PRD and live AI product · Next.js, serverless

github.com/ohmppatel920/course-navigator

- Ran empathy interviews and journey-mapped how students pick courses during shopping period, then wrote the PRD and shipped against it. The product turns Brown's roughly 5,400-section catalog into a decision, pairing constrained suggestions with a board surfacing schedule conflicts and total workload.
- Built the reliability layer before real users touched it. Conflict detection runs deterministically outside the model, the ranker cannot emit a rating that is not in the retrieved data, and the backend reserves inference budget before committing to a call so cost stays bounded under unpredictable usage.

SideOut: Market Analysis and Product Roadmap

June 2026 – Present

Solo build · Python, MediaPipe/OpenCV, FastAPI

github.com/ohmppatel920/sideout

- Ran a feature and price competitive analysis of Hudl, Balltime, and open-source tools against what teams spend, near \$1,200 a year. Ideation sessions with teammates narrowed it to a staged roadmap delivering standalone user value at each release.
- Shipped v1 off that plan: a pose-estimation pipeline extracting jump height, approach velocity, and load timing from one phone video. Fully typed, 87% test coverage, Docker, CI.

Finding Where an LLM Stops Being Reliable

2025

Independent project · Python, ChromaDB, RAGAS

github.com/ohmppatel920/rag-eval-harness

- Built an evaluation system measuring how reliably an LLM answers questions over dense documents, benchmarking six retrieval configurations against a 30-question test set, instrumented so a failure pointed at the stage that caused it.

EXPERIENCE

Eli Lilly & Company

Indianapolis, IN

Data & Software Intern

May – Aug 2026

- Ran discovery across a biopharmaceutical R&D department to find which manual workflow actually cost the most. Translated that into requirements and shipped the internal application that replaced it. Manual effort fell roughly 40%.
- Onboarded 15 users across several drug discovery programs. Tracked adoption after launch and reworked the parts they routed around.
- Worked across engineering, operations, and scientific teams from scoping through deployment, in daily standups with retrospectives resetting priorities each cycle. Shipped an LLM document extraction service validated against real source documents before release.

Yale University, Schools of Engineering & Medicine

New Haven, CT

Engineering Researcher, AMGEN Scholar

May – Aug 2024 & May – Aug 2025

- Two summers of biomedical research at Yale School of Medicine with no defined scope. Chose which questions were worth pursuing and presented results to the lab each summer. Co-author on a manuscript under review at Springer Nature.

Brown University, School of Engineering

Providence, RI

Engineering Researcher, UTRA Scholar

Sep – Dec 2024

- Built and ran the weekly intake and validation process for an ion chromatography study, 150 samples a week over ten weeks, structuring the 1,500 records all downstream analysis relied on.

LEADERSHIP

Brown University Men's Volleyball · Captain

Sep 2023 – Present

- Captained Brown to its first D1AAA national championship. Own opponent scouting and in-game tactical calls, plus travel, budget, and roster logistics.

Brown Journal of Healthcare Advancement · Editor-in-Chief & Director

Sep 2023 – Present

- Direct a student healthcare publication. Lead 8 editorial teams through draft review across three published cycles.